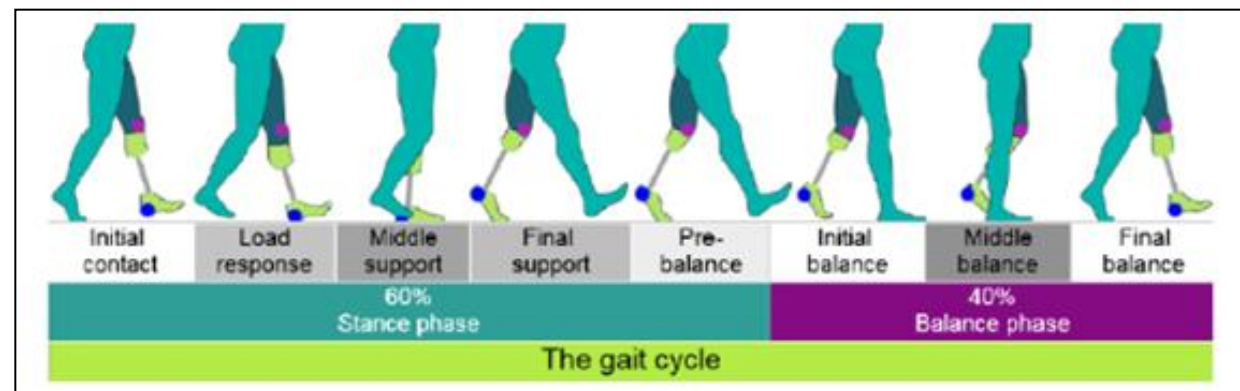
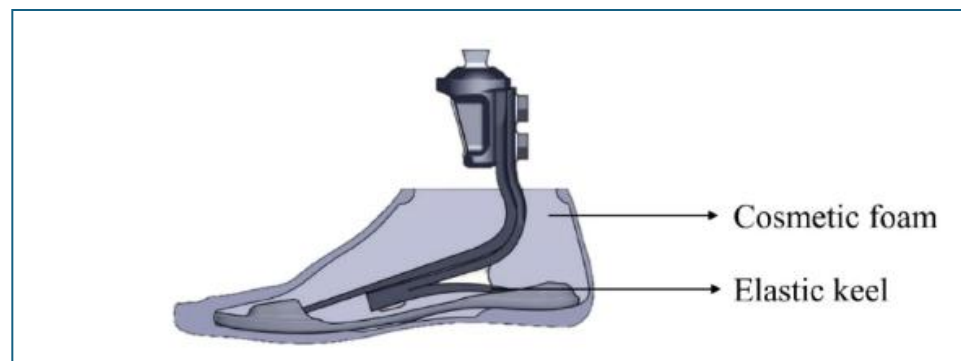
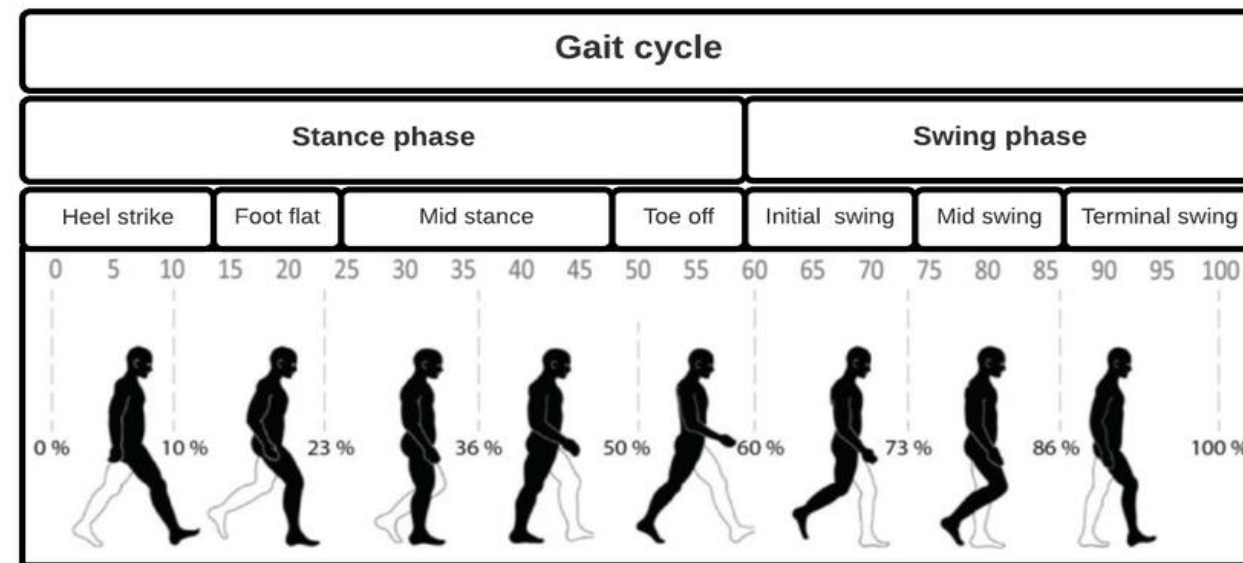
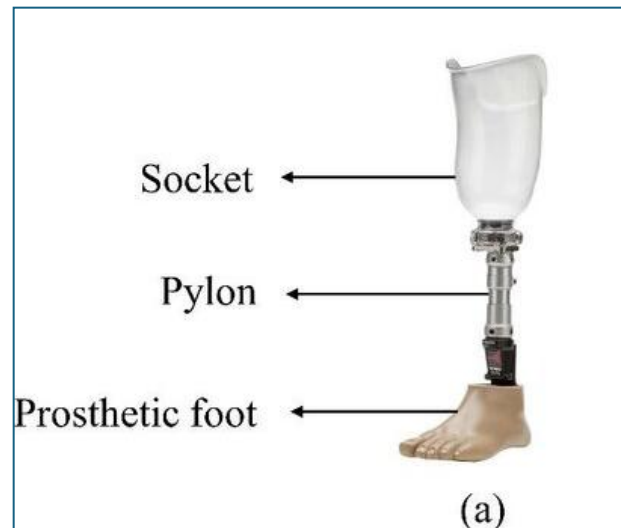
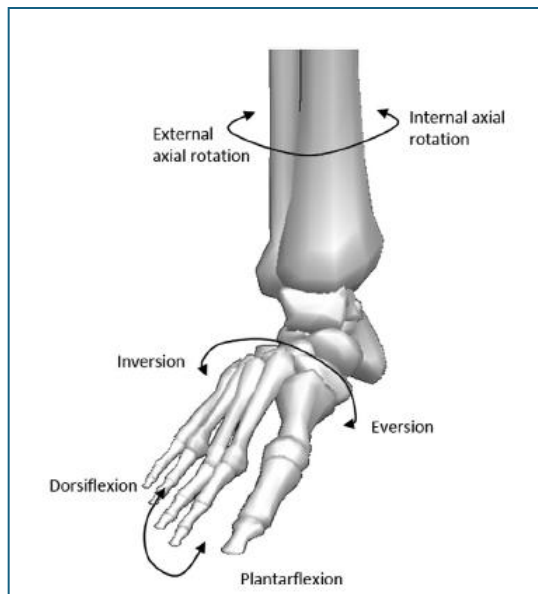


Design and development of a sensorized anklet for restoring proprioception in lower leg

Tesi di Laurea Magistrale in Ingegneria Biomedica

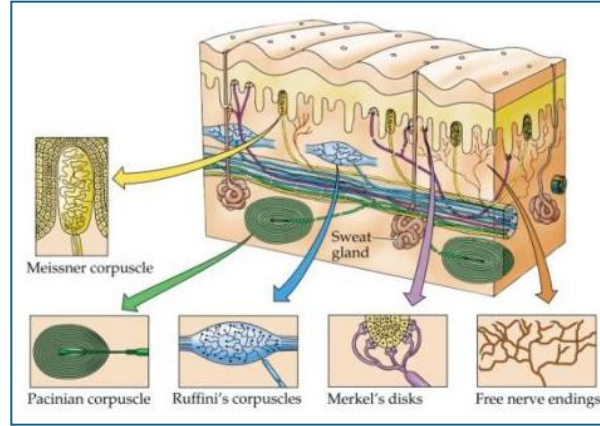
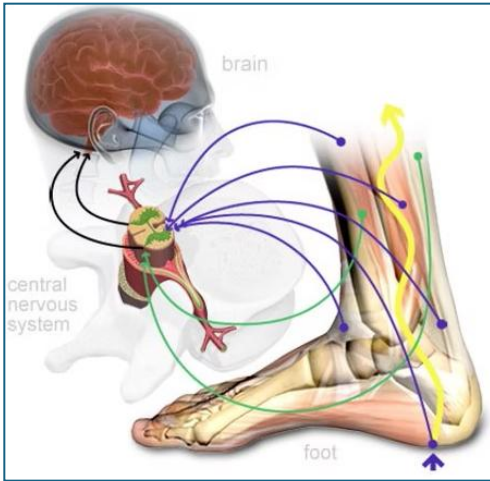
1. Background

Gait cycle in able-bodied and in trans-tibial amputee people



1. Background

Proprioception



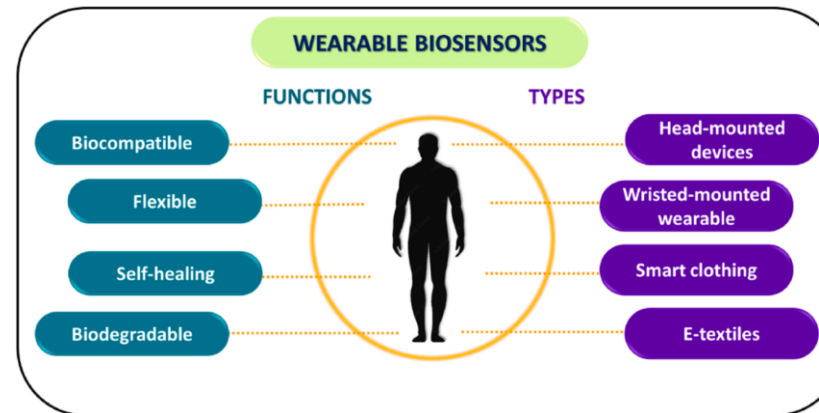
- ability to know body position
- mechanoreceptors in muscles, tendons and skin
- in amputees, auditory, sensitive and visual feedback

Vibrotactile sensory feedback



- small actuators which provide various stimuli to the user

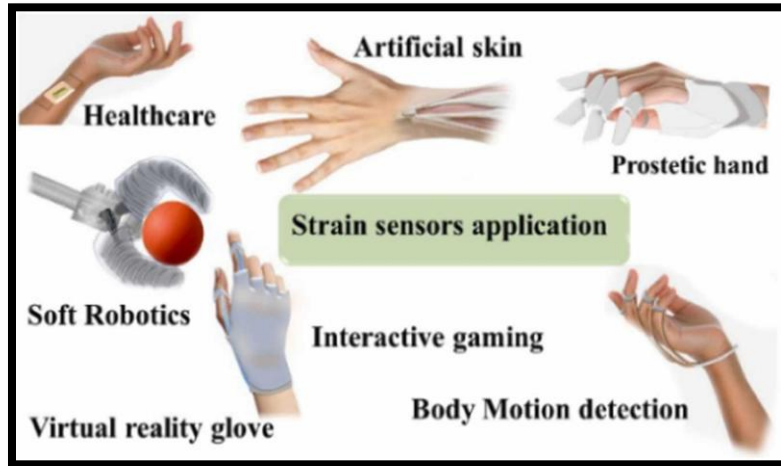
Wearable sensors



- portable devices worn on the body
- continuous and real-time monitoring of vital signals

2. The sensor

Strain sensors



- Piezoresistive
- Piezoelectric
- Capacitive

- Flexible and stretchable materials
- Capacity to react both to low and high level of stress
- Capacity to ensure a certain level of comfort and adaptability to various shapes

Materials
used

Procedure

Characterization



$$GF = \frac{\frac{\Delta R}{R_0}}{\varepsilon}$$

Design and development of a sensorized
anklet for restoring proprioception in lower leg

2. The sensor

Fabrication of the sensors: materials used

The silicone: Ecoflex

TECHNICAL OVERVIEW

	Mixed Viscosity (ASTM D-2393)	Specific Gravity (g/cc) (ASTM D-1475)	Specific Volume (cc./in. ³) (ASTM D-1475)	Pot Life (ASTM D-2471)	Cure Time	Shore Hardness (ASTM D-2240)	Tensile Strength (ASTM D-412)	100% Modulus (ASTM D-412)	Elongation at Break % (ASTM D-412)	Die B Tear Strength (ASTM D-624)	Shrinkage (in./in.) (ASTM D-2566)
Ecoflex™ 5	13,000 cps	1.07	25.8	1 min.	5 min.	5A	350 psi	15 psi	1000%	75 pli	<.001 in./in.
Ecoflex™ 00-50	8,000 cps	1.07	25.9	18 min.	3 hours	00-50	315 psi	12 psi	980%	50 pli	<.001 in./in.
Ecoflex™ 00-30	3,000 cps	1.07	26.0	45 min.	4 hours	00-30	200 psi	10 psi	900%	38 pli	<.001 in./in.
Ecoflex™ 00-33 AF	3,000 cps	1.07	26.0	45 min.	4 hours	00-33	200 psi	10 psi	900%	38 pli	<.001 in./in.
Ecoflex™ 00-20	3,000 cps	1.07	26.0	30 min.	4 hours	00-20	160 psi	8 psi	845%	30 pli	<.001 in./in.
Ecoflex™ 00-20 FAST	3,000 cps	1.07	26.0	20 min.	1 hour	00-20	160 psi	8 psi	845%	30 pli	<.001 in./in.
Ecoflex™ 00-10	14,000 cps	1.04	26.6	30 min.	4 hours	00-10	120 psi	8 psi	800%	22 pli	<.001 in./in.

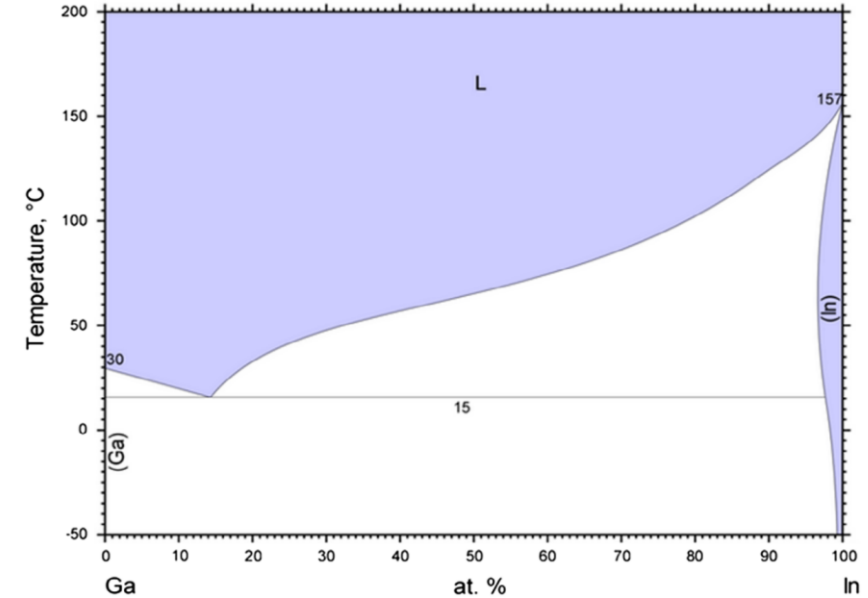
*All values measured after 7 days at 73°F/23°C

Mix Ratio: 1A:1B by volume or weight
Color: Translucent

Useful Temperature Range: -65°F to 450°F (-53°C to 232°C)
Dielectric Strength (ASTM D-147-97a): >350 volts/mil

- Flexibility and deformability
- Low shore hardness
- Large elongation
- Biocompatibility
- Skin-safe

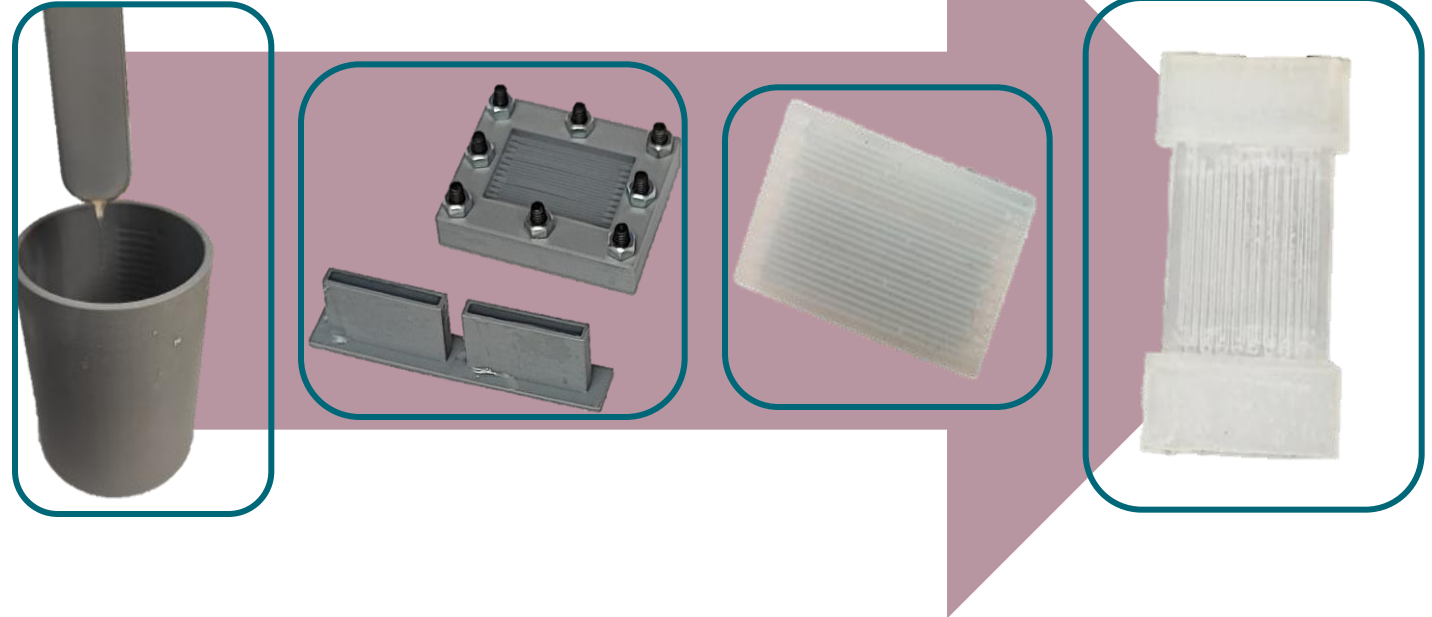
The liquid metal: Gallium-Indium eutectic



- eGaIn is an eutectic alloy (75% Ga and 25% In)
- low toxicity
- good electric conductivity ($3.4 \times 10^6 \text{ S m}^{-1}$)
- melting point at 15.7 °C
- good electrical and thermal properties, unique mechanical, fluidic and superficial characteristics

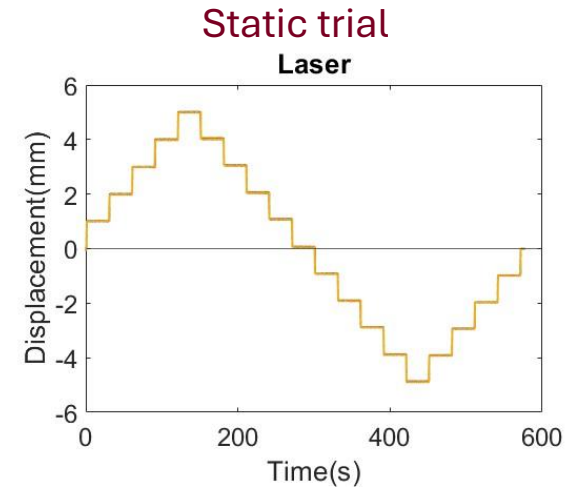
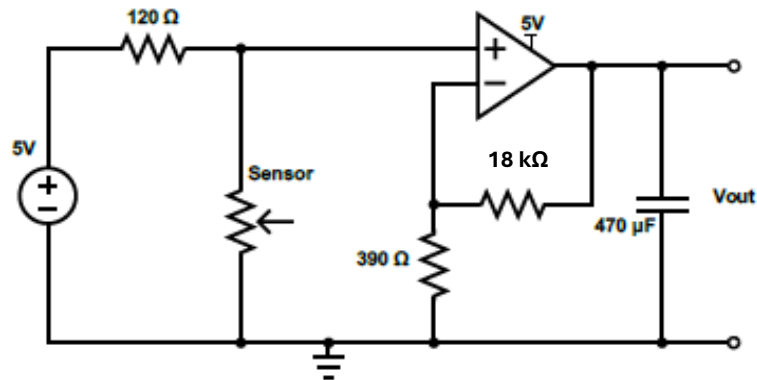
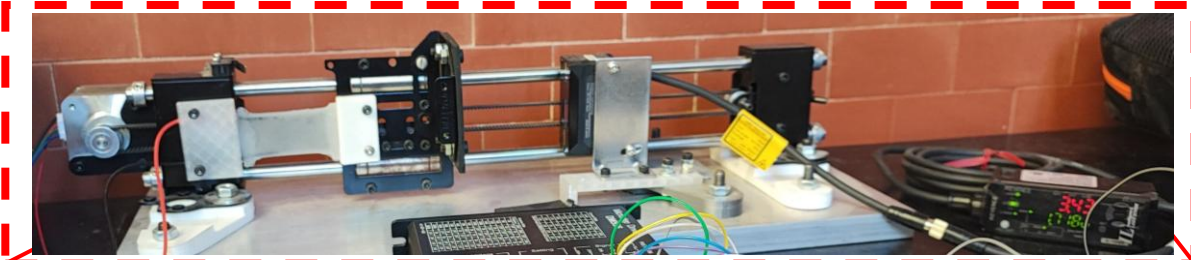
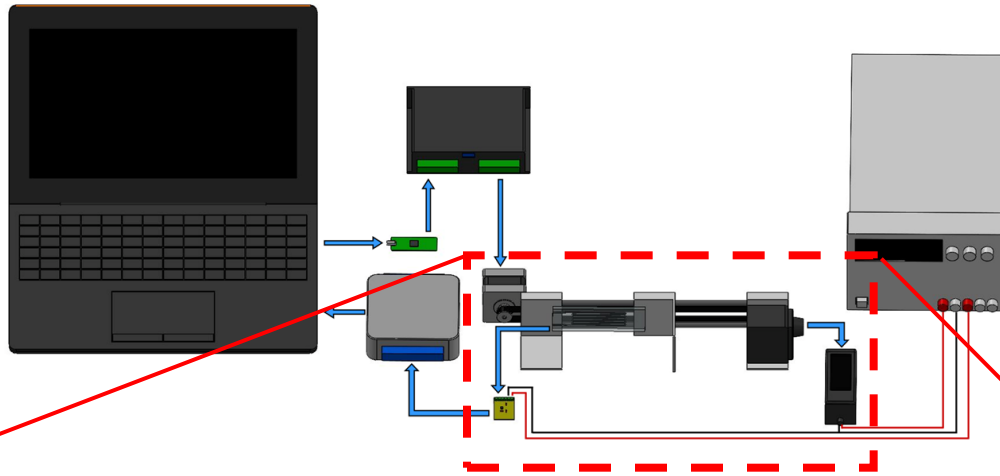
2. The sensor

Fabrication of the sensors: procedure



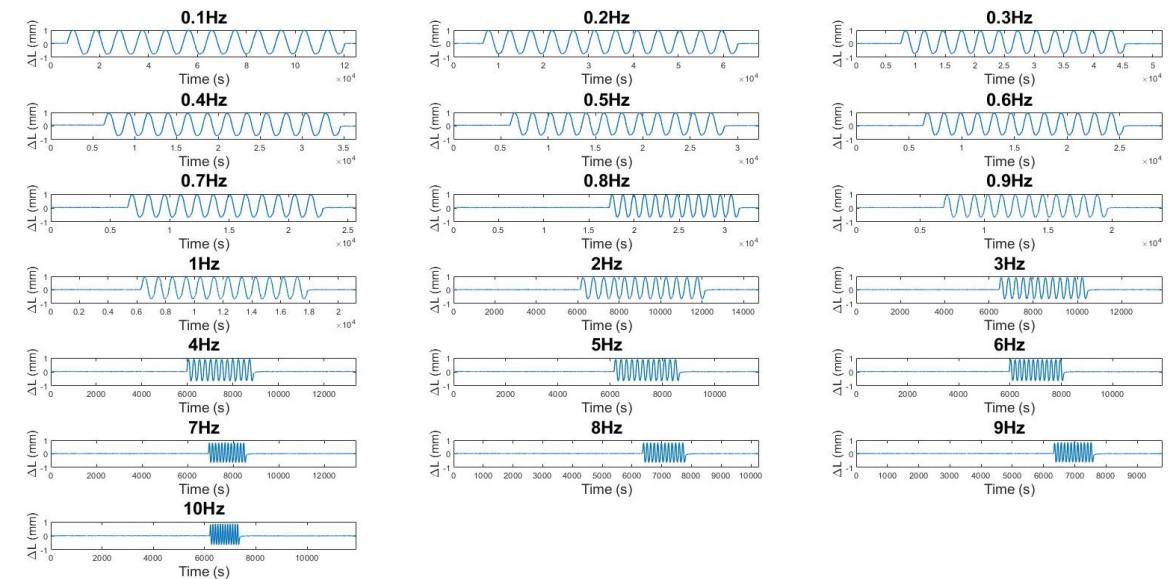
2. The sensor

Characterization of the sensors: setup and procedure



Dynamic trial

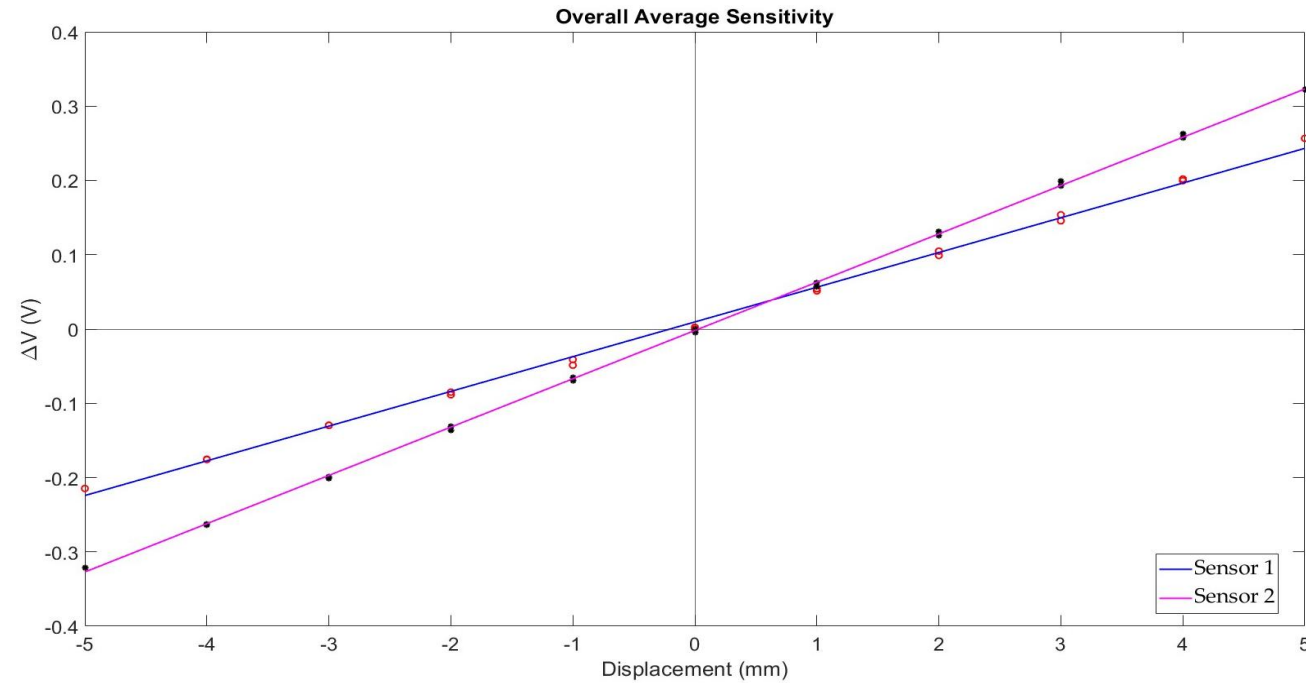
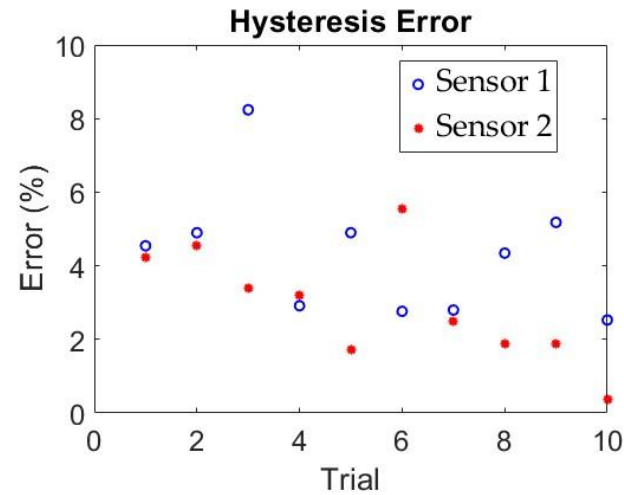
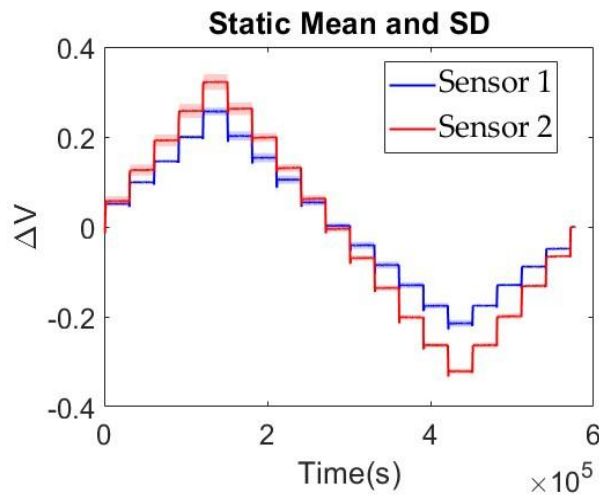
Sinusoidal displacements



Design and development of a sensorized
anklet for restoring proprioception in lower leg

2. The sensor

Characterization of the sensors: static results



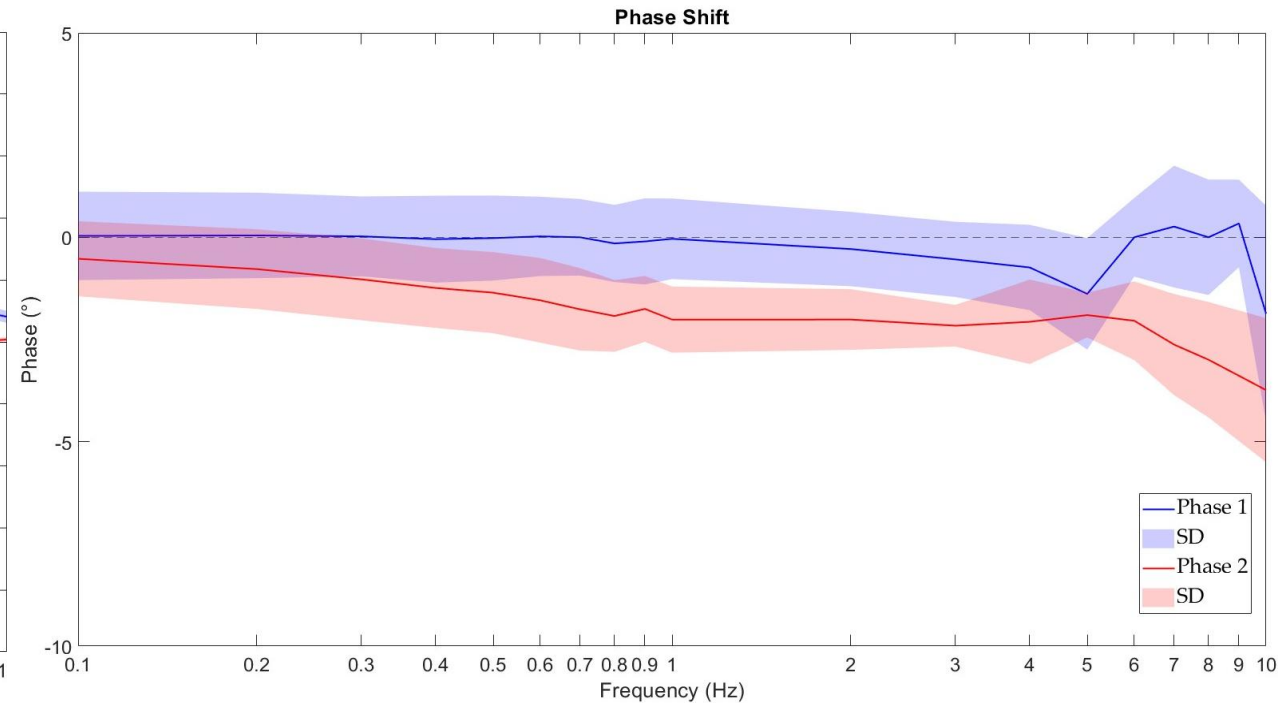
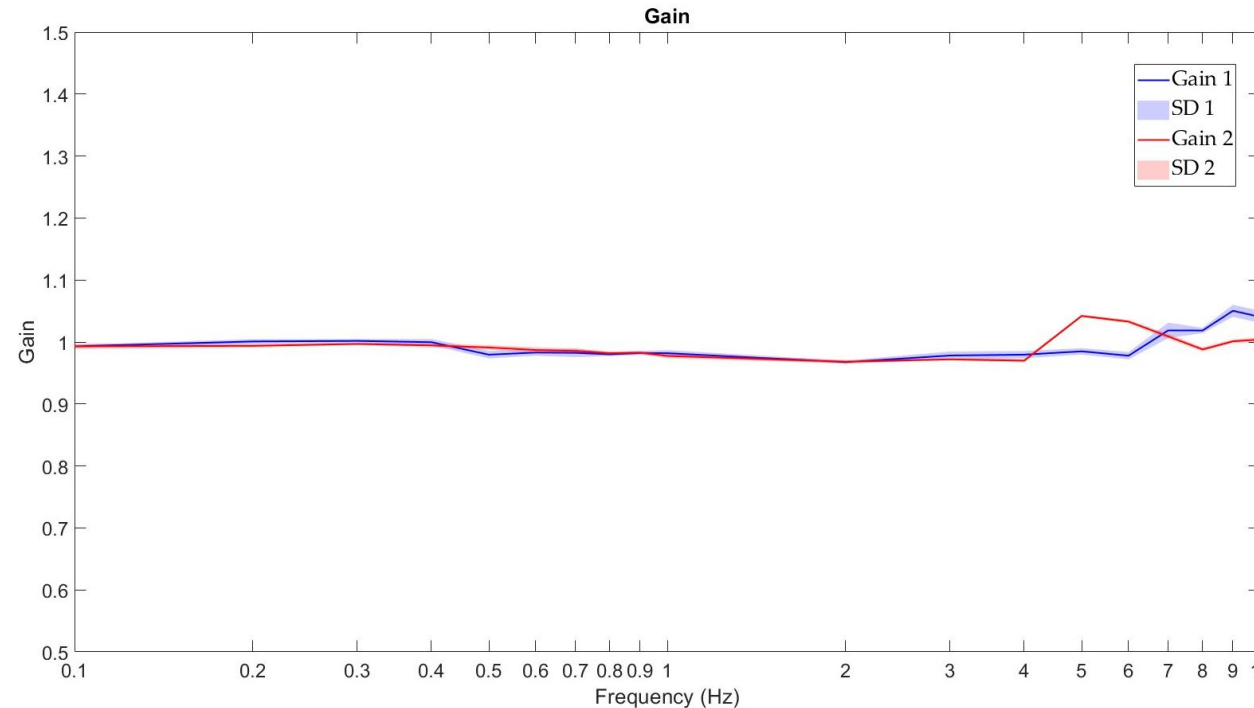
Repeatability (%)		
	Sensor 1	Sensor 2
Average	2,11	2,5
SD	1,26	1,41

	R 2		Sensitivity (mV/mm)	
	Overall 1	Overall 2	Overall 1	Overall 2
Test 1	0,9977	0,9992	46,50	60,93
Test 2	0,9978	0,9995	46,74	62,43
Test 3	0,9937	0,9994	47,76	63,10
Test 4	0,9986	0,9995	46,84	63,47
Test 5	0,9977	0,9996	46,34	67,42
Test 6	0,9984	0,9983	46,84	65,44
Test 7	0,9984	0,9996	46,84	66,22
Test 8	0,9977	0,9997	46,42	66,40
Test 9	0,9975	0,9996	46,30	67,11
Test 10	0,9984	0,9998	46,78	67,65
Average	0,9981	0,9997	46,74	65,02
SD			0,40	2,24

- Both sensors react similarly to the same displacement with a low SD
- At certain steps a slight difference can be noticed
- About hysteresis error, both sensors seem to converge to a lower error value with repeated trials
- Good repeatability
- Both have a linear relationship between displacement and voltage change

2. The sensor

Characterization of the sensors: dynamic results

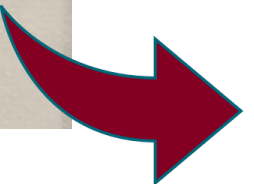
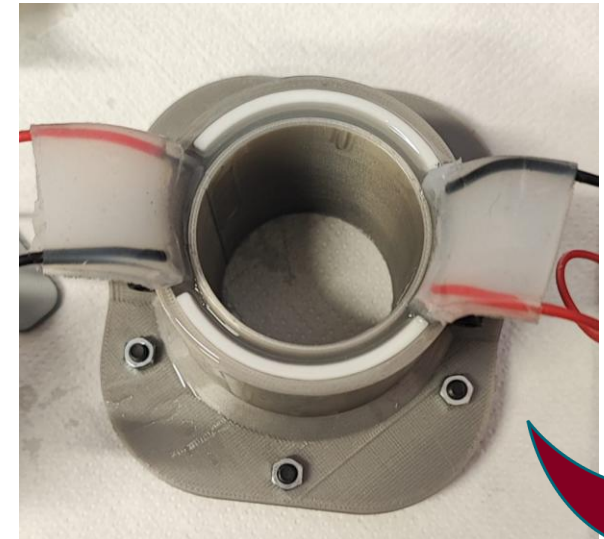
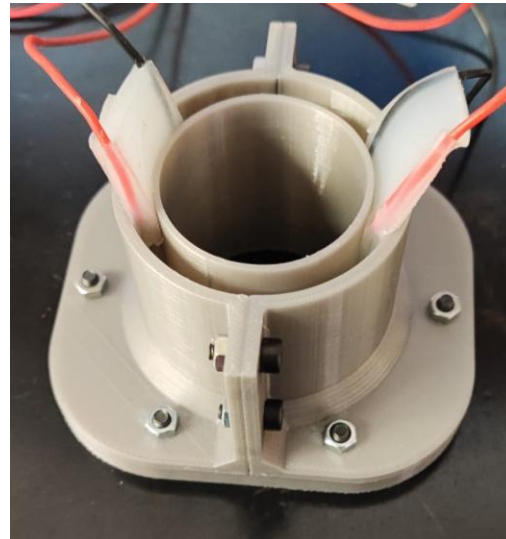
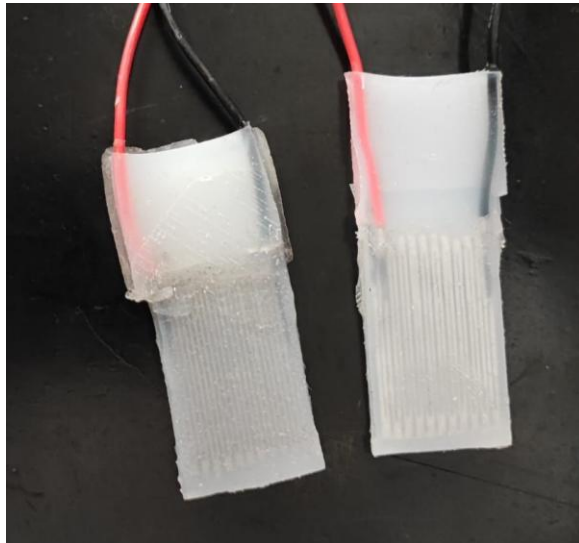
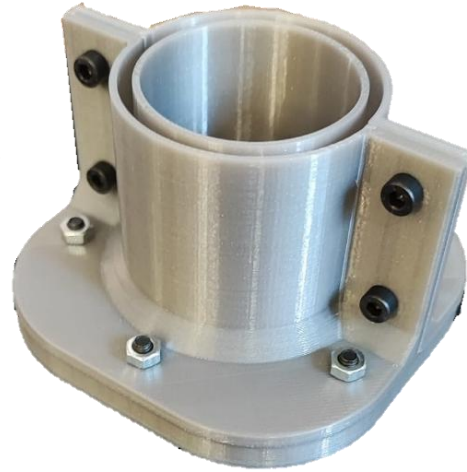
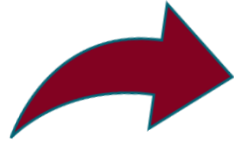
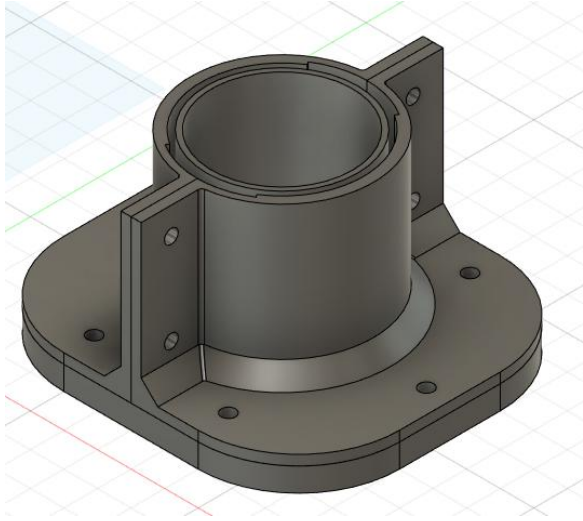


- Both sensors have consistent gain values across the frequency range
- The variability in gain measurements tends to be low
- Above 6 Hz (Sensor 1) and 4 Hz (Sensor 2) a little increase in gain values occurs

- Phase shift tends to decrease with increasing frequency for both sensors
- Delay of Sensor 2 is larger than that of Sensor 1
- Both have similar trend at lower frequencies
- Variability increases at high frequencies

3. The anklet

The fabrication



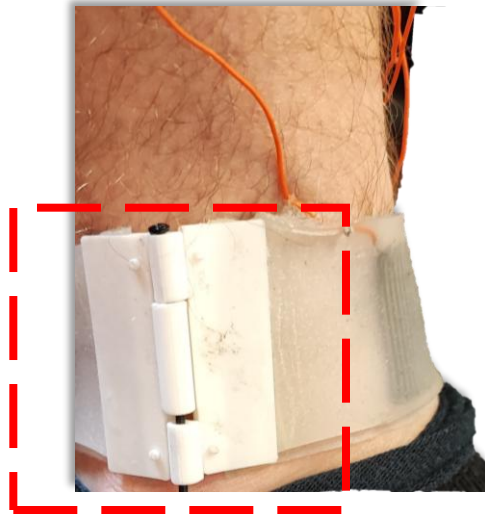
Design and development of a sensorized
anklet for restoring proprioception in lower leg

3. The anklet

The final results

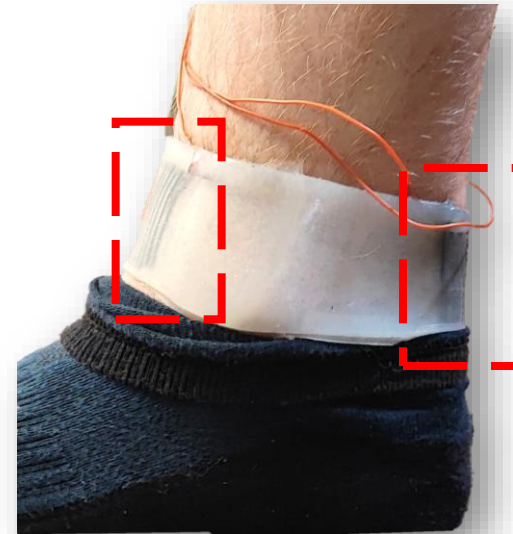


The anklet



The closure of the anklet

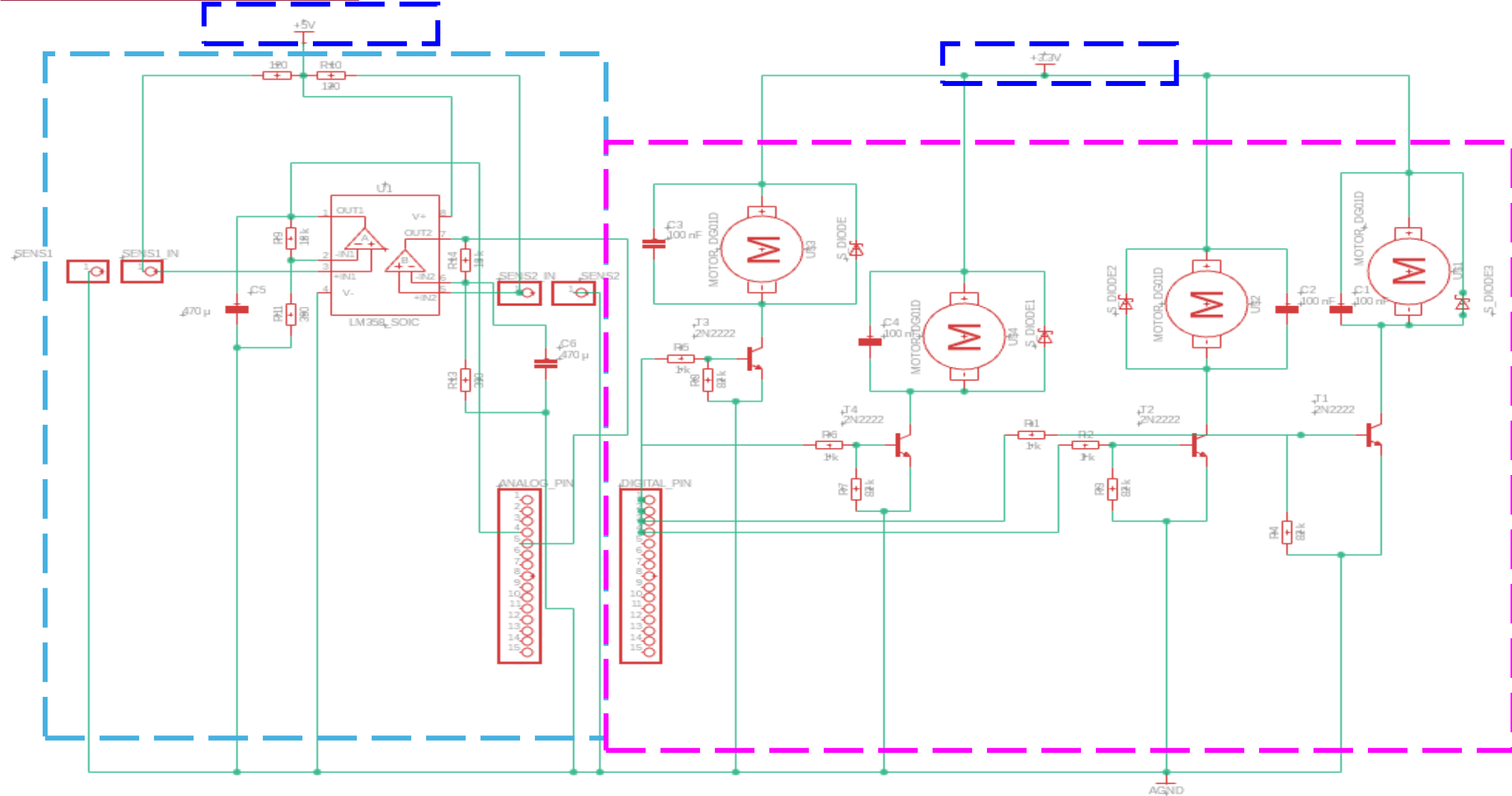
Anklet and sensors: side view



Front view



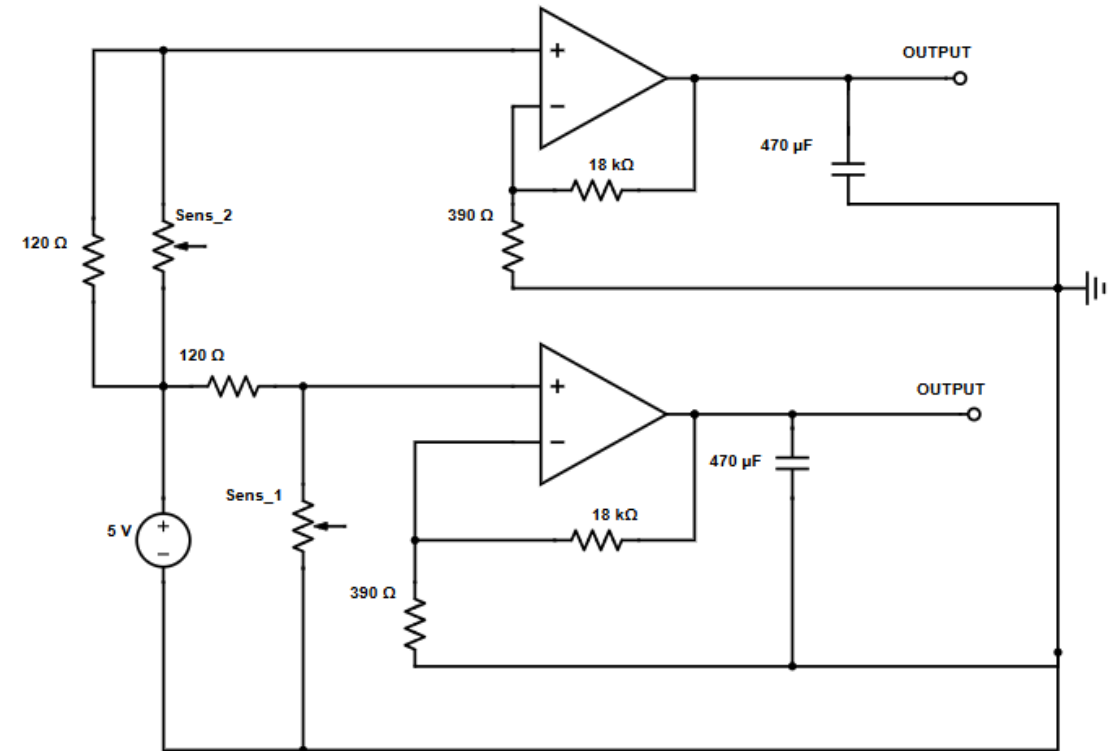
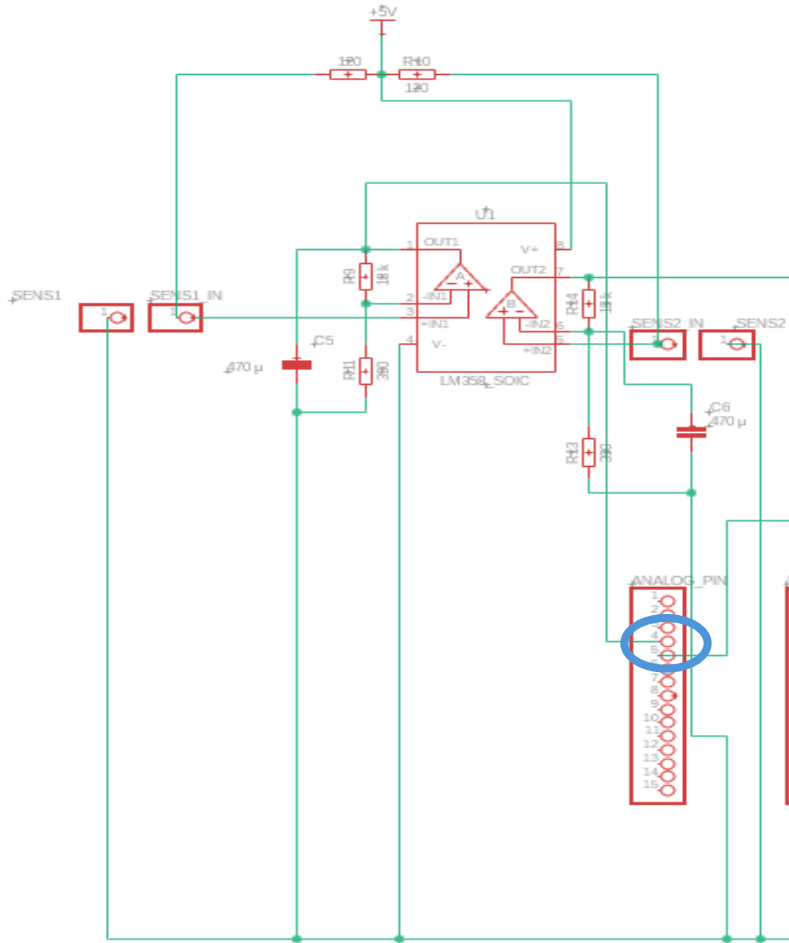
4. The circuit



4. The circuit

Amplification of the sensors

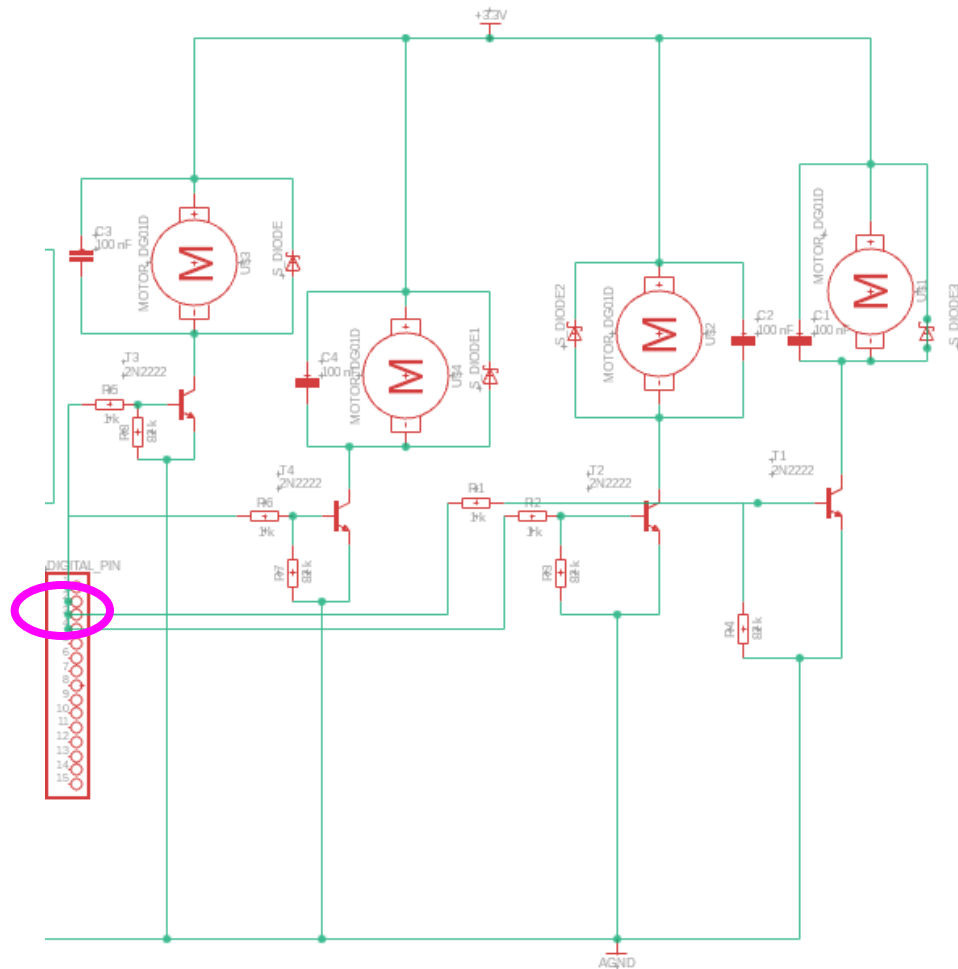
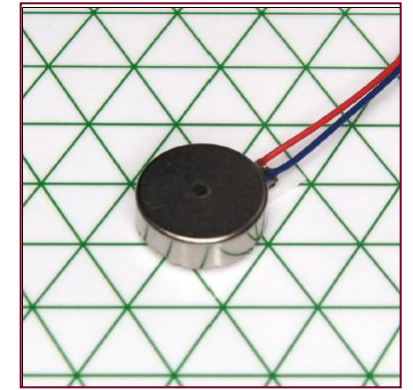
- Operational amplifier (LM258N)
- $120\ \Omega$ resistor (2)
- $18\ \text{k}\Omega$ feedback resistor (2)
- $390\ \Omega$ resistor (2)
- $470\ \mu\text{F}$ capacitor (2)



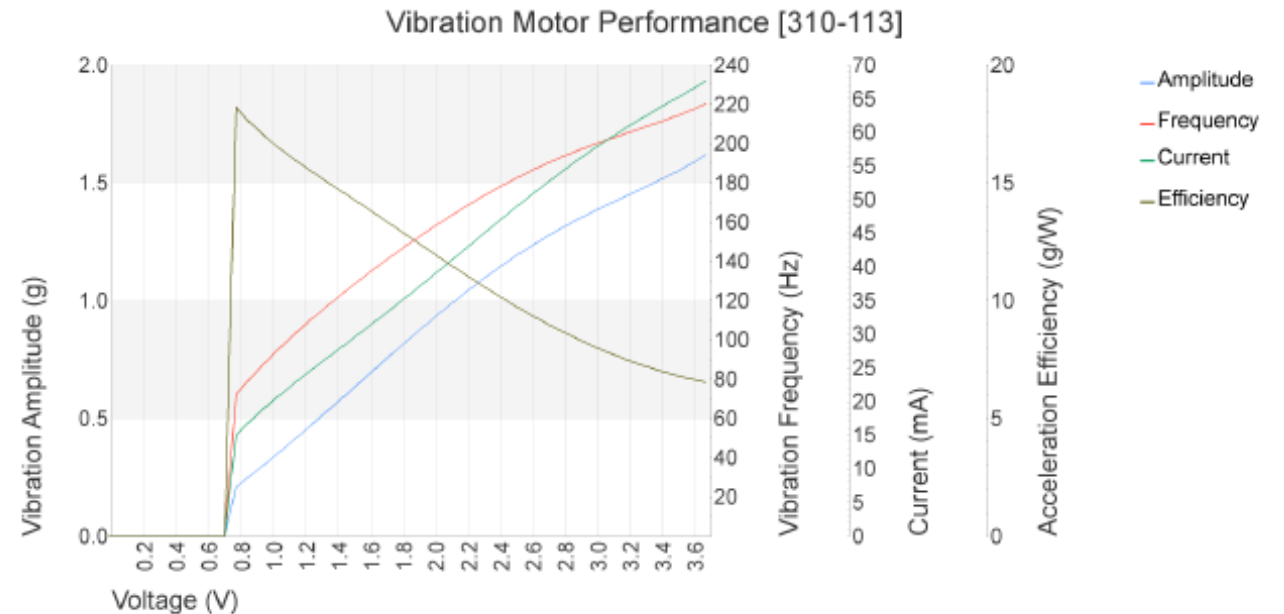
4. The circuit

The ERM motors

- Four ERM motors
- Four 100 nF capacitors
- Four Schottky diodes (1N5819)
- Four BJT (PN2222ATA)
- Four 1 kΩ resistors
- Four 82 kΩ pull-down resistors
- Arduino Nano (digital pin -> **Pulse Width Modulation (PWM)**)



Typical Vibration Motor Performance Characteristics

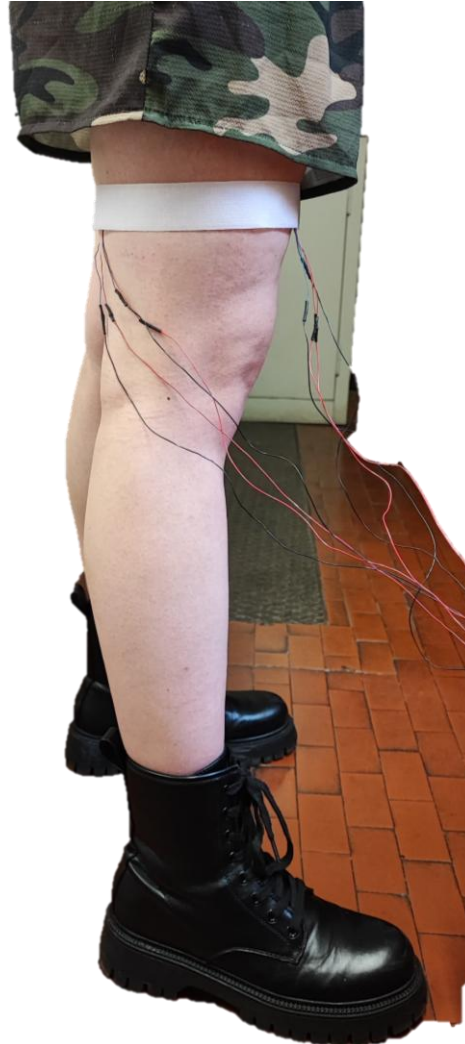


5. The testing of the anklet



- Volunteer A wore the anklet while using a treadmill

- Volunteer B wore a band equipped with motors on the thigh



Setup and procedure

- Volunteer A was involved in the calibration procedure

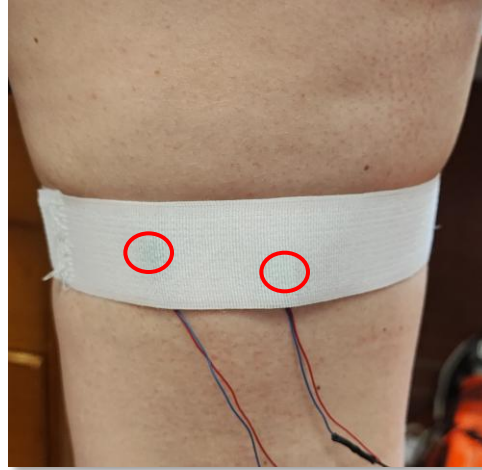
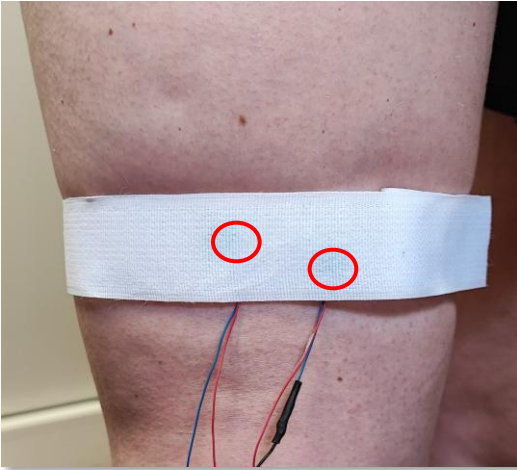
sensorValue = sensorValue1 - sensorValue2

1. *If (sensorValue > 0)*
sensorMax = sensorValue
If (sensorValue) < 1023
sensorMin = sensorValue
2. *If (sensorValue > 0)*
C2Max = sensorValue
If (sensorValue < 1023)
C2Min = sensorValue

- Volunteer B was prepared to not having the possibility to detect movement from volunteer A with sight and hearing senses

5. The testing of the anklet

Setup and procedure



```
if (sensorValue < C2Min)  
    motorPin2 -> C2Min - sensorValue  
    motorPin1 -> 0
```

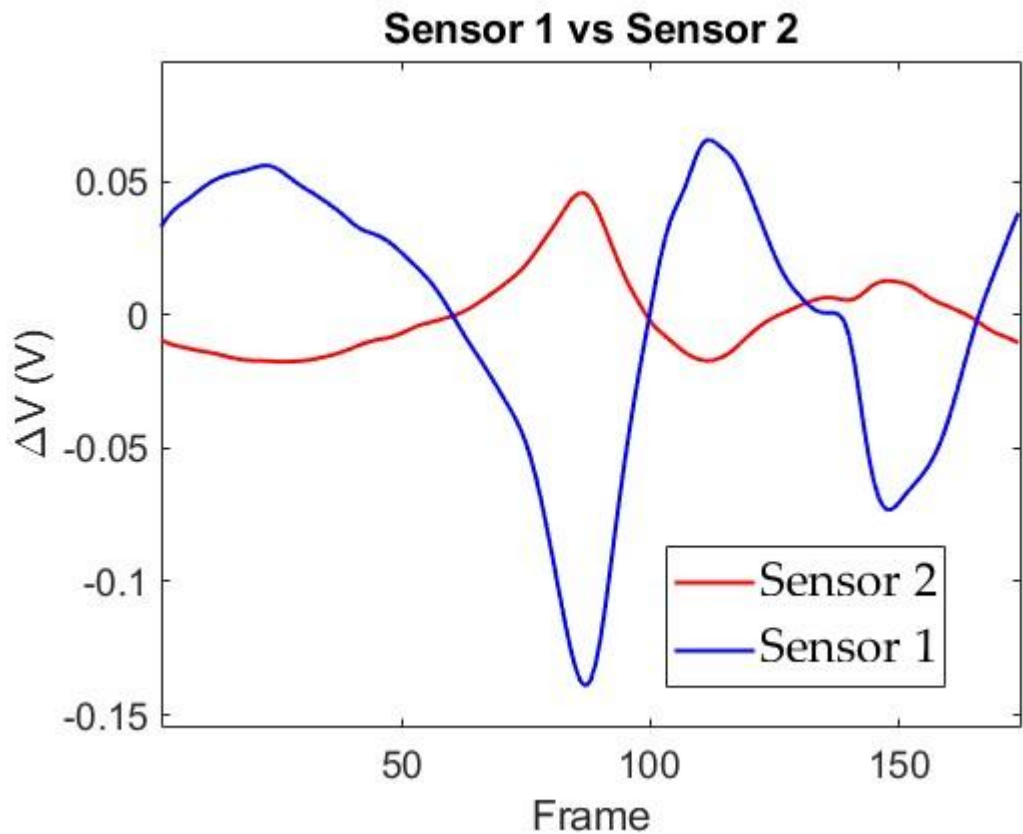
```
if (sensorValue > C2Max)  
    motorPin1 -> sensorValue - C2Max  
    motorPin2 -> 0
```

- Eight trials were conducted, all at 2 m/s, except for the last one (4 m/s)
- Volunteer A counted the steps done with the leg with the ankle
- Volunteer B counted the steps perceived through the vibration



5. The testing of the anklet

The results



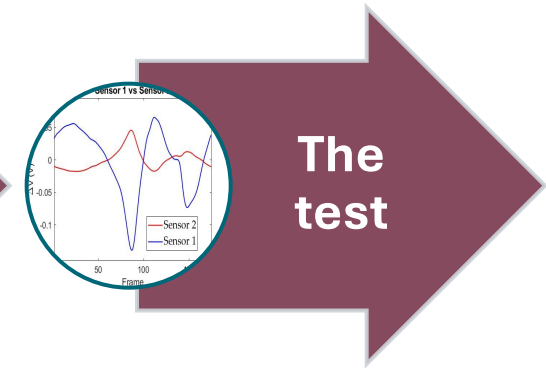
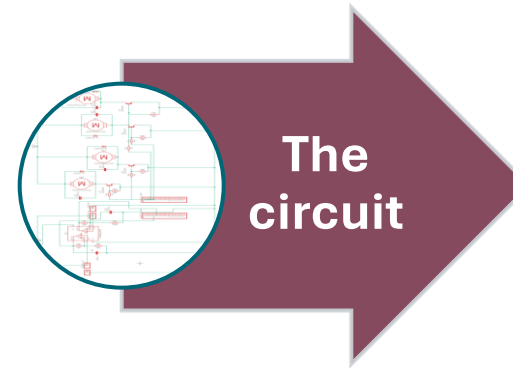
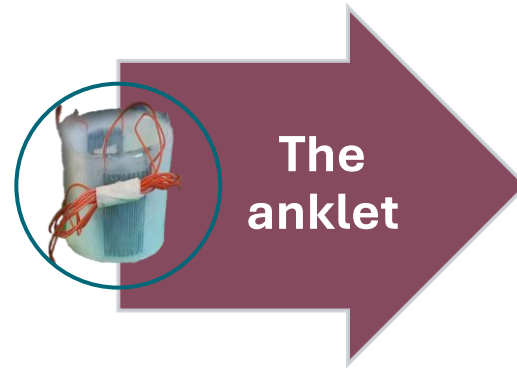
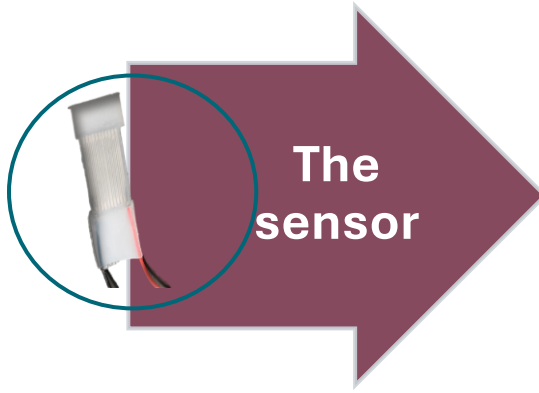
Example of a trial processed with MATLAB

Results of the test			
	Real	Perceived	Percentage
Trial 1	11	18	163,6
Trial 2	24	15	62,5
Trial 3	30	27	90
Trial 4	17	16	94,12
Trial 5	15	15	100
Trial 6	4	4	100
Trial 7	33	28	84,8
Trial 8	14	13	92,9
Average		17	98,5
SD		7,2	27

The overall results of the test

6. Conclusions

Has the aim of this project
been met?



- Good sensitivity and repeatability
- Acceptable hysteresis error
- Satisfactory dynamic response

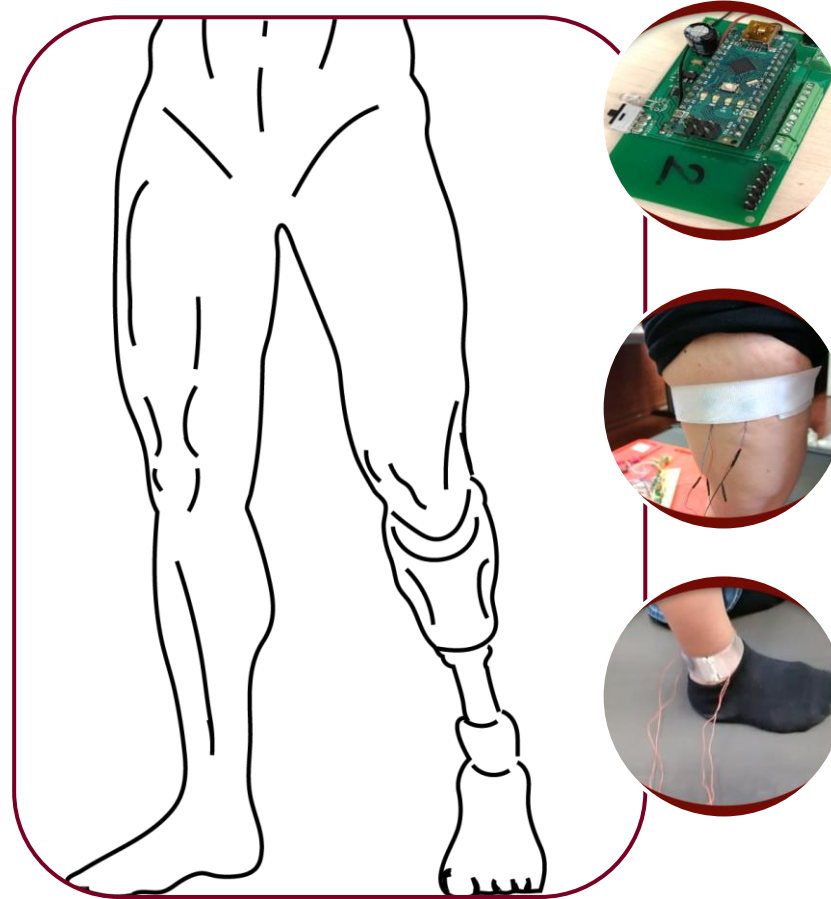
- Successfully equipped with two functioning sensors
- Comfortable and easy to wear

- Able to collect data from the two sensors
- Able to transmit data to vibratile motors

- Increase in the accuracy of the volunteers' responses with practice
- Vibrotactile feedback ensured

7. What's next

- Increase of signal amplification, especially of posterior sensor



PCB

Band with motors

Ankle

